

## Symmetric Group.

Def. Let  $A$  be a non-empty set.

A Group of bijections on  $A$ :

$$S_A \stackrel{\text{def}}{=} \{ f: A \rightarrow A \mid f \text{ is bijection} \}$$

is called a Symmetric Group on  $A$ .

Lemma. Define  $S_n$  as Symmetric group on a set  $\{1, 2, \dots, n\}$ .

If  $|A| = n$ , then  $S_A \cong S_n$ .

Proof. Since  $|A| = n$ , there exists a bijection

$$f: A \rightarrow \{1, 2, \dots, n\}$$

Define a map:

$$Q: S_A \rightarrow S_n; \sigma \mapsto f \circ \sigma \circ f^{-1}$$

$A \rightarrow A \quad A \rightarrow A \quad \{1, \dots, n\} \rightarrow A$

It is well defined, because Composition of bijection is bijection.

And, injectiveness is given by just Cancellation;

$$f \circ \sigma_1 \circ f^{-1} = f \circ \sigma_2 \circ f^{-1} \Rightarrow \sigma_1 = \sigma_2$$

Surjectivity is given by: for any  $\alpha \in S_n$ ,  $f^{-1} \circ \alpha \circ f$  is contained in  $S_A$  and

$$Q(f^{-1} \circ \alpha \circ f) = f \circ f^{-1} \circ \alpha \circ f \circ f^{-1} = \alpha.$$

Finally, Homomorphism because: for any  $\sigma_1, \sigma_2 \in S_A$ ,

$$Q(\sigma_1 \circ \sigma_2) = f \circ \sigma_1 \circ \sigma_2 \circ f^{-1} = f \circ \sigma_1 \circ f^{-1} \circ f \circ \sigma_2 \circ f^{-1} = Q(\sigma_1) \circ Q(\sigma_2).$$

Now, we denote all permutation group as  $S_n$ .

## Cycle Decomposition.

Let  $S_n$  be a Symmetric Group. Then, every element of  $S_n$  can be represented by

$$\sigma = \begin{pmatrix} 1 & 2 & \dots & n \\ a_1 & a_2 & \dots & a_n \end{pmatrix} \text{ where } \{a_i\}_{i=1}^n \text{ is a sequence having range as } \{1, \dots, n\}.$$

Choose  $a \in \{1, \dots, n\}$ , Construct a Cycle: for the smallest positive integer  $k_1 \geq 1$  s.t.  $\sigma^{k_1}(a) = a$ ,

$$(a \ \sigma^1(a) \ \sigma^2(a) \ \dots \ \sigma^{k_1-1}(a))$$

Choose  $b \in \{1, \dots, n\}$  s.t.  $\forall 0 \leq i \leq k_1-1, b \neq \sigma^i(a)$ . Then, similarly, we can construct:

$$(b \ \sigma^1(b) \ \sigma^2(b) \ \dots \ \sigma^{k_2-1}(b))$$

Since  $\{1, \dots, n\}$  is finite set, this process will terminate in finite time.

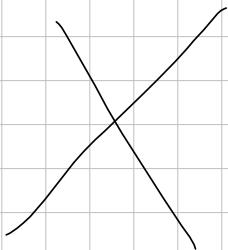
And, the elements in a cycle are mutually distinct, because: if for some  $0 \leq i < j \leq k_1-1$ ,

$$\sigma^i(a) = \sigma^j(a) \Rightarrow \sigma^0(a) = a = \sigma^{j-i}(a)$$

But this contradicts to minimality of  $k_1$ .

Moreover, these two cycles are disjoint: if for some  $0 \leq r \leq k_1-1, 0 \leq t \leq k_2-1$ ,

$$\sigma^r(a) = \sigma^t(b) \Rightarrow b = \sigma^{r-t}(a)$$



Cycle decomposition.

Let  $S_A$  be a symmetric group of a set  $A$ . Let  $\sigma \in S_A$  be given.

Define a relation on  $A$  as:

$$a \sim b \iff a = \sigma^i(b) \text{ for some } i \in \mathbb{Z}.$$

Then, it is an equivalent relation because

$a \sim a$  is trivial, put  $i=0$ .

If  $a \sim b$ , then  $a = \sigma^i(b)$  for some  $i \in \mathbb{Z}$ . Take  $\sigma^{-i}$  to both side, then  $\sigma^{-i}(a) = b$ .

If  $a \sim b$  and  $b \sim c$ , then, for some  $i, j \in \mathbb{Z}$ ,  $a = \sigma^i(b)$  and  $b = \sigma^j(c)$ . Now,  
$$a = \sigma^i(b) = \sigma^i(\sigma^j(c)) = \sigma^{i+j}(c)$$

Therefore, we can consider a partition

$$A/\sim = \{C_a \mid a \in A\} \text{ where } C_a = \{\sigma^i(a) \mid i \in \mathbb{Z}\}.$$

If  $A$  is infinite,  $A$  can have infinitely many infinite orbits.

But, if  $A$  is finite,  $A$  must have finitely many finite orbits.

Regardless of Cardinality of  $A$ , the orbits have the following properties:

Lemma. Let  $A$  be a set, and let  $\sigma \in S_A$  be a permutation on  $A$ .

If  $a \in A$  satisfies that  $\sigma^n(a) = a$  for some  $n \geq 0$ ,

then, for  $p = \min \{n \geq 0 \mid \sigma^n(a) = a\}$ ,

$$\{a, \sigma(a), \dots, \sigma^{p-1}(a)\} = C_a$$

and each  $\sigma^i(a)$  ( $0 \leq i \leq p-1$ ) are mutually disjoint.

Proof. The existence of such  $p$  is given by the Well-ordering principle.

And, since  $\sigma^p(a) = a$ , for any  $k \in \mathbb{Z}$ ,

$$\sigma^k(a) = \sigma^{k \bmod p}(a).$$

Finally, to show that mutually disjointness, suppose that for some  $0 \leq i < j \leq p-1$ ,

$$\sigma^i(a) = \sigma^j(a)$$

Take  $\sigma^{-i}$  to both side, then

$$a = \sigma^{j-i}(a)$$

But it is contradiction to the minimality.  $\square$

If exists, the smallest number  $p \geq 1$  s.t.  $\sigma^p(a) = a$  is called period of  $a$  with respect to  $\sigma$ .

Clearly, if  $C_a$  is finite set, then all elements have a same period.

Def. Let  $\sigma \in S_n$  be a permutation

If  $\sigma$  has at most one cycle having more than one element,  
then  $\sigma$  is called a Cycle, and denote

$$\sigma = (a \ \sigma(a) \ \sigma^2(a) \ \dots \ \sigma^{p-1}(a)) \text{ where } p \geq 1 \text{ is period of } a.$$

Thm. Let  $S_n$  be a Symmetric Group on  $n$  letters.

Every permutation of  $S_n$  can be represented by finite composition of disjoint cycles.  
Moreover, disjoint cycles commute.

Proof. Let  $\sigma \in S_n$  be given. Then, there exist finite orbits of  $\sigma$ :

$$C_{a_1}, \dots, C_{a_k}.$$

For each  $1 \leq i \leq k$ , define:

$$\sigma_i : \{1, \dots, n\} \rightarrow \{1, \dots, n\} : x \mapsto \begin{cases} \sigma(x) & \text{if } x \in C_{a_i} \\ x & \text{if } x \notin C_{a_i} \end{cases}$$

Then, clearly each  $\sigma_i$ 's are Cycles, and commute. Moreover,

$$\sigma = \sigma_1 \circ \dots \circ \sigma_k.$$

Transposition decomposition

Def. Let  $S_n$  be a Symmetric Group on  $n$  letters.

A length of 2 cycle is called a transposition in  $S_n$ .

Thm. Every Cycles in  $S_n$  Can be represented by finite product of transpositions.

Proof. Let a Cycle  $(a_1 a_2 \dots a_k) \in S_n$  be given. Then, by computation,

$$(a_1 a_2 \dots a_k) = (a_1 a_k)(a_1 a_{k-1}) \dots (a_1 a_2)$$

Now, we can consider all permutation in  $S_n$  is a product of transpositions.

Note: transposition decomposition is not unique. for example,

$$(1 2 3) = (1 3)(1 2) = (1 3)(1 2)(1 3)(1 3)$$

Thm. No permutation in  $S_n$  can be represented by both even and odd product of transpositions.

Proof. Let  $\sigma \in S_n$  be given. To use contradiction, Suppose that

$$\sigma = f_1 \circ \dots \circ f_{n_1} = g_1 \circ \dots \circ g_{n_2}$$

where  $f_i, g_i$  are transposition, and  $n_1$  is odd,  $n_2$  is even number.

Consider the standard basis for  $\mathbb{R}^n$ ,

$$\{e_1, \dots, e_n\}$$

clearly,

$$\det(e_1 \dots e_n) = 1.$$

$$\text{And, } \det(e_{f_1(1)} e_{f_1(2)} \dots e_{f_1(n)}) = -1$$

Because it is the change of two rows for  $I_n$ .

$$\text{Inductively, } \det(e_{f_1 \circ \dots \circ f_{n_1}(1)} \dots e_{f_1 \circ \dots \circ f_{n_1}(n)}) = (-1)^{n_1} = -1$$

$$\text{and } \det(e_{g_1 \circ \dots \circ g_{n_2}(1)} \dots e_{g_1 \circ \dots \circ g_{n_2}(n)}) = (-1)^{n_2} = 1.$$

Contradiction.

Thm. Let  $S_n$  be a Symmetric Group on  $n$  letters.

1. Let  $\sigma \in S_n$  is a cycle of length  $m$ . Then

$\sigma^i$  is  $m$ -cycle if and only if  $\gcd(i, m) = 1$

Proof. Denote  $\sigma = (a_0 a_1 \dots a_{m-1})$  as cycle notation. That is,  $\sigma^k(a_i) = a_{i+k \bmod m}$ .

Let  $2 \leq i \leq m-1$  be given.

$\sigma^i$  is  $m$ -cycle  $\Leftrightarrow a_i, \sigma^i(a_i), \sigma^{2i}(a_i), \dots, \sigma^{(m-1)i}(a_i)$  are distinct

$\Leftrightarrow a_i, a_{2i \bmod m}, a_{3i \bmod m}, \dots, a_{mi \bmod m}$  are distinct.

$\Leftrightarrow \forall x \neq y \in \{1, \dots, m\}, x \neq y \Rightarrow xi \not\equiv yi \pmod m$

$\Leftrightarrow \gcd(i, m) = 1$ .

2.  $\sigma \in S_n$  has order 2 iff  $\sigma$  is product of commute transpositions.

Proof. Let  $\sigma \in S_n$  has order 2. That is, for all  $1 \leq i \leq n$ ,  $\sigma^2(i) = i$  and  $i \neq \sigma(i)$ .

Now, a cycle in the cycle must be identity or transposition.

Since order of  $\sigma$  is not 1, it contains at least one transposition.

The converse is trivial.

3. Let  $p$  be a prime number. Then,

$\sigma \in S_n$  has order  $p$  iff  $\sigma$  is product of commute  $p$ -cycles.

4. Let  $\sigma \in S_n$  be a permutation with cycle decomposition

$$\sigma = \sigma_1 \circ \dots \circ \sigma_k.$$

Then,

$$|\sigma| = \text{lcm}(|\sigma_1|, \dots, |\sigma_k|).$$

Note: we already know that the map

$$\text{sgn}: S_n \rightarrow \{\pm 1\}: \sigma \mapsto \begin{cases} 1 & \text{if } \sigma \text{ is even} \\ -1 & \text{if } \sigma \text{ is odd} \end{cases}$$

is well-defined

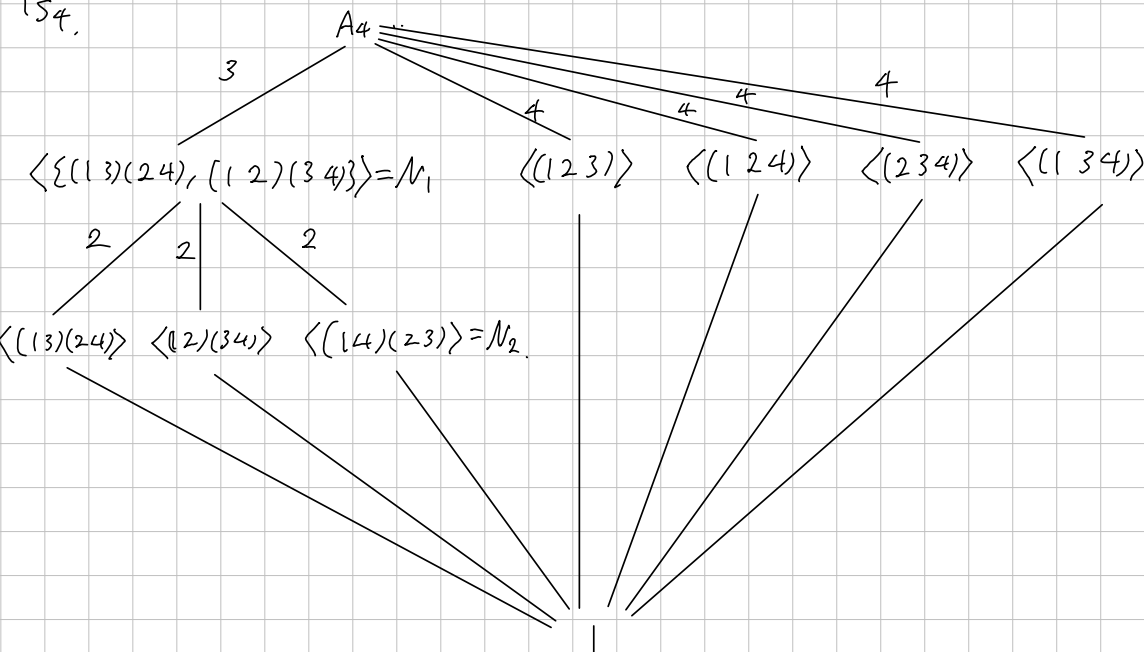
Lemma.  $\text{sgn}: S_n \rightarrow \{\pm 1\}$  is Epimorphism.

Def. Let  $S_n$  be a Symmetric Group on  $n$  letters.

A subgroup  $A_n \stackrel{\text{def}}{=} \ker(\text{sgn})$  is called Alternating Group of degree  $n$ .

Clearly,  $A_n \trianglelefteq S_n$  and  $|S_n/A_n| = 2$ , because  $S_n/A_n \cong \{\pm 1\}$ .

$S_4$ ,



Thm.  $S_4$  is Solvable.

Proof. Since  $N_1$  is subgroup of all elements of order 2, and conjugate cannot change the order,  $N_1 \trianglelefteq A_4$ .

Now,  $1 \trianglelefteq N_2 \trianglelefteq N_1 \trianglelefteq A_4 \trianglelefteq S_4$  all factors above are  $\mathbb{Z}_2$  or  $\mathbb{Z}_3$ , thus abelian.

Thm.

Let  $\sigma, \tau$  be elements of the symmetric group  $S_n$  and suppose

$$\sigma = (a_1, a_2 \dots a_{k_1})(b_1, b_2 \dots b_{k_2}) \dots$$

Then, the conjugate

$$\tau \circ \sigma \circ \tau^{-1} = (\tau(a_1) \tau(a_2) \dots \tau(a_{k_1}))(\tau(b_1) \dots \tau(b_{k_2})) \dots$$

Proof. observe that if  $\sigma(i) = j$ , then

$$\tau \circ \sigma \circ \tau^{-1}(\tau(i)) = \tau(j)$$

Def.

If  $\sigma \in S_n$  is the product of disjoint cycles of lengths  $n_1 \leq n_2 \leq \dots \leq n_r$ , then the integers  $n_1, \dots, n_r$  are called the cycle type of  $\sigma$ .

Prop.

Let  $\sigma_1, \sigma_2 \in S_n$  be given. Then

$\sigma_1$  and  $\sigma_2$  are conjugate if and only if  $\sigma_1$  and  $\sigma_2$  have same cycle type.

Further, the number of conjugacy classes of  $S_n$  equals the number of partitions of  $n$ .

Proof. If  $\sigma_1 = \tau \sigma_2 \tau^{-1}$  for some  $\tau$ , then the theorem above gives that they have same cycle structure. Conversely, suppose  $\sigma_1, \sigma_2$  have same cycle type.



Thm.  $A_5$  is simple group.  
prwf.